École Normale Supérieure de Paris

Licence de Biologie L3

Année 2022–23

Mathématiques I : ce qu'un biologiste ne doit pas ignorer

Examen

The test lasts 2 hours. Individual course notes are allowed. The use of any electronic device is forbidden except a calculator.

1 Dynamics of a Leslie population

We consider a population structured in n age classes. For $1 \leq k \leq n$, we note $x_k(t)$ the number of individuals of the class k at the generation t and $x(t) = [x_1(t) \dots x_n(t)]^\top$ the vector describing the whole population at this same generation. We suppose that the evolution of this population is governed by the recurrence

$$x(t+1) = A\, x(t) \tag{1}$$

where A is the Leslie matrix

$$A = \begin{bmatrix} f_1 & f_2 & \cdots & \cdots & f_n \\ s_1 & 0 & \cdots & \cdots & 0 \\ 0 & \ddots & \ddots & & \vdots \\ \vdots & \ddots & \ddots & \ddots & \vdots \\ 0 & \cdots & 0 & s_{n-1} & 0 \end{bmatrix}$$

where all coefficients f_i and s_i are assumed to be strictly positive. We note moreover

$$\ell_1 = 1 \qquad \text{et} \qquad \ell_k = \prod_{i=1}^{k-1} s_i \quad \text{pour } 2 \leq k \leq n.$$

Characteristic polynomial.

1. Interpret the coefficients f_i and s_i.

2. Let $B_{k1} \in \mathcal{M}_{k-1}$ and $B_{k2} \in \mathcal{M}_{n-k}$ defined by :

$$B_{k1} = \begin{bmatrix} s_1 & -\lambda & & \\ & \ddots & \ddots & \\ & & \ddots & -\lambda \\ & & & s_{k-1} \end{bmatrix}, \qquad B_{k2} = \begin{bmatrix} -\lambda & & & \\ s_{k+1} & \ddots & & \\ & \ddots & \ddots & \\ & & s_{n-1} & -\lambda \end{bmatrix}.$$

Denoting $0_{p,q}$ the matrix $p \times q$ whose elements are all zero, calculate the determinant of the matrix $B_k \in \mathcal{M}_{n-1}$:

$$B_k = \begin{bmatrix} B_{k1} & 0_{k-1,n-k} \\ 0_{n-k,k-1} & B_{k2} \end{bmatrix}.$$

3. Show that the characteristic polynomial of A is

$$P_A(\lambda) = (f_1 - \lambda)(-\lambda)^{n-1} + \sum_{k=2}^{n} (-1)^{k-1} f_k \ell_k (-\lambda)^{n-k}.$$

4. Deduce that the largest eigenvalue in modulus, denoted λ_1, of the matrix A verifies

$$\sum_{k=1}^{n} \ell_k f_k \lambda_1^{-k} = 1.$$

Eigenvectors. We are now interested in the left and right eigenvectors of the matrix A. We note

$$a = \sum_{k=1}^{n} \ell_k \lambda_1^{-k}, \qquad b = \sum_{k=1}^{n} k\, \ell_k f_k \lambda_1^{-k}.$$

5. Show that the vector v with coordinates

$$v_k = \frac{1}{a} \ell_k \lambda_1^{-k}, \qquad 1 \leq k \leq n,$$

is an right-eigenvector of A associated with the eigenvalue λ_1.

6. Show that the vector u with coordinates

$$u_k = \frac{1}{b v_k} \sum_{j=k}^{n} \ell_j f_j \lambda_1^{-j}, \qquad 1 \leq k \leq n,$$

is an left-eigenvector of A associated with the eigenvalue λ_1.

Asymptotic behavior. Finally, we are interested in the asymptotic behavior of the vector $x(t)$ describing the composition of the population after t generations.

7. Calculate $\sum_{k=1}^{n} v_k$ and $\sum_{k=1}^{n} v_k u_k$.

8. Starting from an initial composition vector $x(0)$, what is the asymptotic behavior of $x(t)$ when t tends to infinity?

2 Linear tangent application to a quadratic form

Consider a symmetric matrix $A \in \mathcal{M}_n$, a vector $v \in \mathbb{R}^n$ and the function

$$\begin{array}{rccl} f: & \mathbb{R}^n & \mapsto & \mathbb{R} \\ & x & \to & f(x) = x^\top A x + v^\top x. \end{array}$$

1. Show that there exists a vector $g(x) \in \mathbb{R}^n$, which you will specify, such that the linear application tangent to f at x is

$$\begin{array}{rccl} D_x f: & \mathbb{R}^n & \mapsto & \mathbb{R}^n \\ & h & \to & D_x f(h) = g(x)^\top h. \end{array}$$

2. Assuming that A is invertible, determine the stationary point x^* where $g(x)$ is zero.

3. Give a condition on A for x^* to be a local minimum. (We may calculate the hessian matrix of A).

4. Discuss the usefulness of the assumption that A is symmetric.

3 Third order differential system

Consider the nonlinear ordinary differential equation

$$\dot{x} = f(x) = -x^3 + 7x^2 - 14x + 8. \tag{2}$$

1. Determine the set of stationary points of (2).

2. Study the stability of each of these stationary points.

3. Draw the stability diagram of the system.

4. From a system behavior point of view, what are the respective roles of each of the stationary points.

4 Markov chain with two absorbing states

We consider a Markov chain $(X_n)_{n \geq 0}$ with values in $\{0, \ldots, \ell\}$, and we note $p_{i,j}$ the transition probability from state i to state j. It is assumed that

$$p_{0,0} = p_{\ell,\ell} = 1$$

and that, for any $1 \leq i, j \leq \ell - 1$,

$$p_{i,0} = a > 0, \qquad p_{i,\ell} = b > 0, \qquad p_{i,j} \geq 0.$$

We also note $T_{i,j}$ the time to reach the state j from the state i:

$$T_{i,j} = \min\{n : X_n = j \mid X_0 = i\}.$$

1. Give the transition matrix and draw the transition graph of this Markov chain.
2. What are the communication classes of this chain? Give their nature. What are the possible asymptotic behaviors of this chain?
3. Let $1 \leq i \leq \ell - 1$, give the probability that $X_{n+1} \in \{1, \dots \ell - 1\}$ knowing that $X_n = i$.
4. Deduce that, for $1 \leq i \leq \ell - 1$, noting $c = a + b$,

$$\mathbb{P}\{T_{i,0} = n\} = a(1-c)^{n-1}, \qquad \mathbb{P}\{T_{i,\ell} = n\} = b(1-c)^{n-1}.$$

5. For $1 \leq i \leq \ell - 1$, we now note T_i the absorption time (in 0 or in ℓ) :

$$T_i = \min(T_{i,0}, T_{i,\ell}).$$

   Show that T_i has a geometric distribution $\mathcal{G}(c)$: $\mathbb{P}\{T_i = n\} = c(1-c)^{n-1}$.
   Deduce its expectation.
6. For $1 \leq i \leq \ell - 1$, calculate the probability that, starting from $X_0 = i$ the Markov chain reaches the state 0 before the state ℓ.
7. What are the respective expectations of $T_{i,0}$ and $T_{i,\ell}$ for $1 \leq i \leq \ell - 1$?